

Hammad Ali

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EDUCATION

Bahauddin Zakariya University
Bachelor of Science, Computer Science

Multan, PK
2023 - 2027

SKILLS - LANGUAGES & FRAMEWORKS

- | | | |
|--------------|-------------|--------------|
| ▪ JavaScript | ▪ Node.js | ▪ React |
| ▪ Rest API | ▪ Langchain | ▪ Postgresql |
| ▪ Redis | ▪ Git | ▪ Github |

PROJECTS

Emerald AI

Full Stack AI chat assistant

- Designed and implemented secure authentication using JWT stored in httpOnly, SameSite cookies with bcrypt password hashing, enforcing strict per-user data isolation across all session and message endpoints to prevent cross-account access.
- Built a Redis-backed rate-limiting layer using rate-limiter-flexible with a deliberate fail-open design, ensuring legitimate users remain unaffected even if the Redis cache becomes temporarily unreachable.
- Architected the backend using an MVC pattern across Node.js, Express, and PostgreSQL, with parameterized queries and centralized error handling applied consistently across every controller to align with OWASP Top 10 security practices.
- Integrated Groq's Llama 3 via LangChain for AI-generated chat responses, with persistent chat sessions and message history stored in PostgreSQL.
- Developed a responsive React frontend with Zustand for state management and optimistic UI updates, allowing sent messages to render instantly while the AI response is generated server-side.

AI Interview Bot

Frontend React Application

- Engineered a voice-first interview experience using the Web Speech API, enabling real-time speech recognition with continuous mic restart logic to ensure uninterrupted answer capture across all five interview questions.
- Built a keyword-based scoring engine as a client-side fallback, evaluating candidate answers against role-specific and level-specific keyword sets across six job categories (Frontend, Backend, Data Science, Product Manager, UI/UX, HR) with structured 0-9 scoring.
- Designed an animated interviewer interface featuring a centered WebGL-style orb with CSS-based glow and pulse effects synchronized to speech synthesis state, replacing a traditional chat UI with an audio-first human interview feel.
- Implemented a persistent score history system using localStorage, tracking performance across multiple attempts per role and level, with a dashboard-style report card displaying progress bars, attempt history, and grade breakdowns.
- Wrote 15 unit and interaction tests using Vitest and React Testing Library, covering component rendering, user interaction flows, edge cases including skipped answers, and score calculation correctness.

Ecommerce API

RESTful API for an e-commerce platform

- Built a production-grade REST API for a full e-commerce flow - product catalog, shopping cart, checkout, and order history - using Node.js, Express, and PostgreSQL with an MVC architecture.
- Implemented JWT authentication via httpOnly cookies with role-based access control, restricting product and inventory management to admin users while keeping cart and checkout actions scoped per customer.
- Designed a relational schema separating orders from order items, preserving the price at the time of purchase so historical orders remain accurate even after a product's price changes.

- Hardened the API against common attack vectors per OWASP guidelines: parameterized queries to prevent SQL injection, Helmet.js for secure headers, rate limiting on authentication and global endpoints, and strict CORS origin whitelisting.
- Added input validation and sanitization with express-validator, centralized error handling with no stack-trace leakage in production, and structured logging via Winston.